



Department of Education

**STEM Education
National Schools of Excellence
Teacher's Guide
Grade 12
Term 1**

**Science
Technology
Engineering
Mathematics**

Issued free to schools by the Department of Education

Published in 2021 by the Department of Education, Papua New Guinea

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ISBN

Acknowledgements

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. It was written by writers from universities, secondary schools, new university graduates with expertise in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus. This document was developed with support from the General Education Services Division of the Department of Education

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SECRETARY’S MESSAGE

This Engineering Teachers Guide is to be used exclusively by teachers of Engineering to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Engineering Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Engineering before undertaking the advanced concepts introduced in this syllabus.

The Teachers Guide conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this Teachers Guide, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Engineering as a fundamental field of science.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Engineering is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Engineering to human lives, including good health and environment.

In order to ensure Teachers, discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Engineering to be used for the Grade 11 STEM Education Program in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

The Engineering Syllabus introduced to the Schools of Excellence as a critical component of the STEM Education is mostly concerns

Teaching Engineering requires laboratory sessions as an essential part of the teacher guide. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Engineering.

The Grade 11 Engineering Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Engineering is a highly specialized subject students undertaking Engineering are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Engineering through the process of imparting basic Engineering knowledge and skills to student

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Agricultural Biotechnology	1. Farming system in Russia - Krasnodar Krai 2. Agricultural machinery for farming 3. Agricultural Engineering - Preparing Land for Agriculture 4. Designing Markham for farming – Agricultural Engineering
HEALTH AND MEDICINE	12.1 Biomedical Engineering	1. Introduction to biomedical engineering 2. Disciplines of Biomedical Engineering 3. Applications of Biomedical Engineering 4. Diagnosis of Dengue
ENERGY	12.1 Energy Generation	1. Solar Energy – Solar Power (Photovoltaic System) 2. Wind Energy 3. Hydro-Electricity Energy (Hydropower) 4. Geothermal Energy
BATTERY TECHNOLOGY	12.1 Building Batteries	1. Efficiency of Batteries 2. Lead-Acid Batteries 3. Lithium-Ion Batteries 4. Future of Battery
ARTIFICIAL INTELLIGENCE	12.1 Constructing Intelligence	1. Introduction to Artificial Intelligence 2. Fundamentals of Artificial Intelligence 3. Natural Language Processing 4. Future of Artificial Intelligence
ROBOTICS	12.1 Fundamentals	1. Robotics – The Perfect Human Beings? 2. Robotics – Hardware 3. Robotics – Software

	of Robotics Engineering	4. Constructing a robot
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RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

The Engineering was introduced as a Bridging Course to the Schools of Excellence. The syllabus introduced the students to the Engineering Domain and expects them to develop an appreciation of the discipline in a broader context. In this teacher guide students are introduced to advanced concepts in Engineering as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Teachers Guide, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Teachers Guide. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domain

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM ENGINEERING CONTENT OVERVIEW – GRADE 12, TERM 1

Table 1.1: Sequencing & Benchmarking STEM Engineering Content Standards

Setting STEM Engineering Content Standards for Progressive Learning and Assessment National STEM Assessment			
DOMAIN: Engineering			
Thema BIOTECHNOLOGY		Broad Content Standard:	
Module 11.1: Biomedical Engineering		12.1. Assessing the importance of biotech crops and engineering simple techniques involved in farming for developing countries compared to developed countries.	
Topic	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Biomedical Engineering	12.1.1a: Understand and define biotech crops in relation to agricultural biotechnology 12.1.1b: Evaluate the significance of biotech crops 12.1.1c: Identify and discuss how engineering is applied 12.1.1d: Interact and work together as a team to develop people skills..	Student's ability to master knowledge and skills of; - Define Biotech crops as an aspect of agricultural biotech - Evaluate the significance of biotech crops - Analyze and asses the developments of farming in Russia - Evaluate and comment on farming in developing countries – PNG	Practical Assignment + Quiz + Test + Case Study
2. Disciplines of Biomedical Engineering	12.1.2a: Discuss and analyze the operation and farming in Russia 12.1.2b: Asses on the irrigation, drainage systems and location of villages/households 12.1.2c: Articulate the engineering techniques involved		

3. Applications of Biomedical Engineering	12.1.3a: Critically comment and assess on selection of land for agriculture 12.1.3b: Discuss on the procedures involved in preparing land for agriculture 12.1.3c: Apply strategies and tools for preparing land		
4. Diagnosis of Dengue	12.1.4a: Describe and discuss on designing mass farming of biotech crops 12.1.4b: Analyze and comment on the related procedures and concepts involved 12.1.4c: Critically describe the need for developing biotech farms in PNG 12.1.4d: Develop an understanding on the signification of biotech crops for starch production		

Thema	HEALTH AND MEDICINE	Broad Content Standard:	
Module 11.1: BIOMEDICAL ENGINEERING		12.1. Develop a fundamental knowledge on the applications of biomedical engineering in Medicine.	
Topic	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Biomedical Engineering	12.1.1a: Define biomedical engineering and its related terms. 12.1.1b: Articulate the significance of biomedical engineering 12.1.1c: Analyze the strategies and engineering tools employed by biomedical engineers 12.1.1d: Describe and discuss the different disciplines of biomedical engineering 12.2.1e: Outline the medical significance of the disciplines of biomedical engineering	Students' ability to master knowledge and skills of: - Define and discuss what biomedical engineering and its concepts - Distinguish the different types of Biomedical engineering in medical applications - Evaluating and assessing the different application of biomedical engineering	Practical Assignment + Quiz + Test + Case Study
2. Medical Instrument	12.1.2a: Describe and discuss the purpose of medical equipment. 12.1.2b: Articulate and explain the steps involved in the instrumentation system 12.1.2c: Outline the classifications of biomedical instruments 12.1.2d: Asses the processes involved in the design process of medical instruments		
3. Applications of Biomedical Engineering	12.1.3a: Assess the process involved in the applications of biomedical engineering 12.1.3b: Asses the engineering tools employed in the processes of the applications of biomedical engineering		

4. Case Studies of Biomedical Engineering Applications	12.1.4a: Clearly define and explain foot drop and the engineering solutions to it. 12.1.4b: Understand and analyze the process involving engineering and medicine for the solution 12.1.4c: Discuss and assess different engineering solutions to medical problems that are impending		
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Table 1.2: Mapping Content Progression by Modules

Thema/Module	Content Standard	12.1	12.1
BIOTECHNOLOGY / 12.1 Bioengineering	12.1. Assessing the importance of biotech crops and engineering simple techniques involved in farming for developing countries compared to developed countries.	✓	
HEALTH AND MEDICINE / 12.1 Biomedical Engineering	12.1. Develop a fundamental knowledge on the applications of biomedical engineering in Medicine		✓

THEMA: BIOTECHNOLOGY

MODULE 12.1: AGRICULTURAL BIOTECHNOLOGY

4-5 weeks

The use of biotechnology in agriculture is focused on stable development of agricultural production, solving of food security and safety issues, obtainment of high-quality and ecologically clean food products, agricultural production waste recycling, soil fertility recovery.

PNG's agricultural growth and biotechnology in PNG Biotechnology has the ability to solve problems in PNG's agriculture production and processing in all areas. Plant breeding is used to increase and stabilize yields, improve tolerance to pests, diseases, and abiotic stresses such as drought and cold, and improve food nutritional content.

In this module, it will give a few backgrounds on the farming system in Russia, the machinery needed for farming and how engineering contributes to farming/ engineering agriculture in PNG and an example on how we can cultivate our land mass

Linking Previous Module (STEM Proper)

Pre-requisite:

- Module 11.1/Introduction to Biotechnology- Engineering
- Module 11.1/Topic 2: Industrial Biotechnology -Biology

Broad Content Standard

12.1. Assessing the importance of engineering simple engineering techniques involved in farming for developing countries compared to developed countries.

Content Benchmark

Student's ability to master knowledge and skills of;

- Understand and evaluate the importance of engineering in farming
- Evaluate the significance of machinery used in farming
- Analyze and appreciate the importance of mass farming
- Analyze and assess the developments of farming in Russia
- Evaluate and comment on farming in developing countries – PNG

Topic 1: Farming system in Russia - Krasnodar Krai

Performance Indicators

Students are expected to:

12.1.1a: Understand that Krai is a leading production of Commercial agriculture in Russia

12.1.1b: Identify and discuss More land is diverted into agriculture

12.1.1c: Analyze and interpret Agriculture controls the industry's development in Krasnodar

12.1.1d: Identify and discuss how engineering is applied

12.1.1e: Interact and work together as a team to develop people skills.

Activity 1: Group Discussion

Title: Farming System

- Ask students to discuss have a discussion on the topic and prepare question for them.

Resources: NIL

(Refer to TRB)

Activity 2: Research Activity

Title: Farming System in USA

- Ask students to do research and discuss on the research activity.

Resources: NIL

(Refer to TRB)

Topic 2: Agricultural machinery for farming

Performance Indicators

Students are expected to:

12.1.2a: Understand that machinery for farming is essential

12.1.2b: Understand that machinery for farming is essential

12.1.2c: Discuss the importance of machinery in farming

12.1.2d: Articulate the engineering techniques involved

Activity 1: Exercises

Title: Machinery for Farming

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Activity 2: Research Question

Title: Farming Machinery

- Ask students to do research and discuss on the research activity.

Resources: NIL

(Refer to TRB)

Topic 3: Engineering Agriculture in PNG -Preparing Land for Agriculture

Performance Indicators

Students are expected to:

12.1.3a: Critically comment and assess on selection of land for agriculture

12.1.3b: Discuss on the procedures involved in preparing land for agriculture

12.1.3c: Apply strategies and tools for preparing land farming.

Activity 1: Exercise

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Activity 2: Open Discussion and Project

Title: Preparing Land for Farming

- Ask students to do research and discuss on the research activity.

Resources: NIL

(Refer to TRB)

Activity 3: Practical

Do a simple demonstration with the proposed ideas for preparing of land in your school and suggest the type of crops you think is best for the soil type and location. Include in your discussions:

- Why you have Chosen the type of crop to farm
- Why there is a need for drainage system if any
- How to maintain the soil quality
- Tools and machinery used if any

Resources: NIL

(Refer to TRB)

Topic 4: Designing Markham for farming – Agricultural Engineering

Performance Indicators

Students are expected to:

12.1.4a: Describe and discuss on designing mass farming of biotech crops

12.1.4b: Analyze and comment on the related procedures and concepts involved

12.1.4c: Critically describe the need for developing biotech farms in PNG

12.1.4d: Develop an understanding on the signification of biotech crops for starch production

Activity 1: Project

Title: Agricultural Engineering

- Prepare questions on for students to understand about the topic and give an assignment to have ground their knowledge on the topic on their own.

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Agricultural Biotechnology	12.1. Assessing the importance of engineering simple engineering techniques involved in farming for developing countries compared to developed countries.	1.	Written Quiz Written Quiz Project + Written Test

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1	Written Quiz	Explicitly define Artificial Intelligence and other engineering-related terms.	2.5
		Articulate and explain the contrast of the different categories and types of Artificial Intelligence.	2.5
		Adequately display a desire to become subject matter experts by addressing any real-life issues with logical reasoning before coding.	5
		Clearly explain in their own words the fundamental process of AI development.	5
		Clearly differentiate the branches of AI	2.5
		Explicitly define Artificial Intelligence and other engineering-related terms.	2.5
Subtotal			20
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	30
		Correctly answers short answer questions	40
		Subtotal	80
Module TOTAL			100

THEMA: HEALTH AND MEDICINE

MODULE 12.1: BIOMEDICAL ENGINEERING

4-5 weeks

In this module, the application of engineering principles and problem-solving techniques in the field of medicine will be discussed. It also focuses on improving the health care for human beings at all level. An introduction to the module followed by the disciples and application of biomedical engineering will be discussed in the content.

Linking Previous Module (STEM Proper)

Pre-requisite:

- Module 11.1/Introduction to Engineering (STEM BRIDGING)
- Module 11.2/Introduction to Engineering Design Process
- Module 11.1/Engineering in Health and Medicine

Broad Content Standard

12.1. Develop a fundamental knowledge on the applications of biomedical engineering in Medicine

Content Module Benchmark

Students' ability to master knowledge and skills of:

- Define and discuss what biomedical engineering and its concepts
- Distinguish the different types of Biomedical engineering in medical applications
- Evaluating and assessing the different application of biomedical engineering

Topic 1: Introduction to Biomedical Engineering

Performance Indicators

Students are expected to:

12.1.1a: Define biomedical engineering and its related terms.

12.1.1b: Articulate the significance of biomedical engineering

12.1.1c: Analyse the strategies and engineering tools employed by biomedical engineers

12.1.1d: Describe and discuss the different disciplines of biomedical engineering

12.2.1e: Outline the medical significance of the disciplines of biomedical engineering

Activity 1: Case Study - Title: Prosthesis

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Activity 2: PNG Case Study - Title: Car Accident

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Activity 3: Group Discussion - Title: More Biomedical Engineers Needed

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Activity 4: Class Exercise

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Topic 2: Medical Instrument

Performance Indicators

Students are expected to:

12.1.2a: Describe and discuss the purpose of medical equipment.

12.1.2b: Articulate and explain the steps involved in the instrumentation system

12.1.2c: Outline the classifications of biomedical instruments

12.1.2d: Assess the processes involved in the design process of medical instruments

Activity 1: Classroom Exercise - Medical instrument

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

Topic 3: Applications of Biomedical Engineering

Performance Indicators

Students are expected to:

12.1.3a: Assess the process involved in the applications of biomedical engineering

12.1.3b: Asses the engineering tools employed in the processes of the applications of biomedical engineering

Activity 1: Classroom discussion - Title: Research and Find Out Engineering Solutions

- Prepare questions on for students to understand about the subtopic.

Resources: NIL

(Refer to TRB)

Topic 4: Case Studies of Biomedical Engineering Applications

Performance Indicators

Students are expected to:

12.1.4a: Clearly define and explain foot drop and the engineering solutions to it.

12.1.4b: Understand and analyze the process involving engineering and medicine for the solution

12.1.4c: Discuss and assess different engineering solutions to medical problems that are impending

Class Activity: Group Discussion - Research and Find Out Engineering Solutions

- Prepare questions on for students to understand about the topic.

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Biomedical Engineering	12.1. Develop a fundamental knowledge on the applications of biomedical engineering in Medicine	1.	Written Quiz Written Quiz Project + Written Test

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1	Written Quiz	Explicitly define Artificial Intelligence and other engineering-related terms.	2.5
		Articulate and explain the contrast of the different categories and types of Artificial Intelligence.	2.5
		Adequately display a desire to become subject matter experts by addressing any real-life issues with logical reasoning before coding.	5
		Clearly explain in their own words the fundamental process of AI development.	5
		Clearly differentiate the branches of AI	2.5
		Explicitly define Artificial Intelligence and other engineering-related terms.	2.5
Subtotal			20
Module 12.1 (All four Topics)	Written Test (Multiple Choices, Short Answers, Calculations, etc)	Correctly answers all long questions	10
		Correctly answers all multiple choice	30
		Correctly answers short answer questions	40
		Subtotal	80
Module TOTAL			100

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules	Total Marks	% Weight of Modules
	6		
Theory Knowledge & skills – Test	Test (50)	/50	20
Application of knowledge – Assignments and projects	Assign.1 (30)	/30	30
End of Term Exam	(25 marks)	/25	50
Total marks for term one			100 marks

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Name s	Assessment Task								Total Score (/300)	Percentage Weight (%)				Achievement Level [VHA, HA, SA, LA]
		Tests		Assignments				Project	Exam (/50)		T (20%)	A + P (30%)	Exam (50%)	Ttl CA	
		1	2	1	2	3	4	1							
1	Bill Joe	45	47	20	21	22	23	55	45	278	18.4	25.88	45	89.28	VHA
2	Cara Den	20	25	15	16	17	18	25	30	166	9	18.22	30	57.22	SA
3	Lea May	9	15	5	6	7	8	30	25	105	4.8	9.10	25	38.9	LA

Assessment Reporting

Thema / Biotechnology / Module 12.1: Protein Sequencing Broad Content Standard for Assignments / project 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Determine Protein Sequence (15 marks)	Prepare Lab Report (5 marks)	Answers all Relative Questions to protein sequencing (5 marks)	25 marks
Very High Achievement	Accurately determine protein sequence (13-15)	Accurately writes a lab report with correct sections (5)	Accurately answers all relative questions to protein sequencing (5)	(21-25)
High Achievement	Partially determine protein sequence (10-12)	Partially writes a lab report with correct sections (4)	Partially answers all relative questions to protein sequencing (4)	(16-20)
Satisfactory Achievement	Fairly determine protein sequence (7-9)	Fairly writes a lab report with correct sections (3)	Impartially answers all relative questions to protein sequencing (3)	(11-15)
Low Achievement	Poorly determine protein sequence (4-6)	Poorly writes a lab report with correct sections (2)	Poorly answers all relative questions to protein sequencing (2)	(6-10)
Below minimum Achievement	Needs improvement in determining protein sequence (0-3)	Needs improvement in writing a lab report with correct sections (1)	Needs improvement in answering all relative questions to protein sequencing (1)	(0-5)

Thema Biotechnology / Module 12.1: Protein Sequencing				
Broad Content Standards For Test: 12.1 To determine the amino acid sequence of a protein				
Performance Criteria	Correctly answers all long questions (25 marks)	Correctly answers all multiple choice (10 marks)	Correctly use concepts to design and propose solutions(15 marks)	50 marks
Very High Achievement	Accurately answers all long questions (21-25)	Accurately answers all multiple choice (9-10)	Accurately use concepts to design and propose solutions (13-15)	(41-50)
High Achievement	Partially answers all long questions (16-20)	Partially answers all multiple choice (7-8)	Partially use concepts to design and propose solutions (10-12)	(31-40)
Satisfactory Achievement	Fairly answers all long questions (11-15)	Fairly answers all multiple choice (5-6)	Fairly use concepts to design and propose solutions (7-9)	(21-30)
Low Achievement	Poorly answers all long questions (6-10)	Poorly answers all multiple choice (3-4)	Poorly use concepts to design and propose solutions (4-6)	(11-20)
Below minimum Achievement	Needs improvement in answering all long questions (0-5)	Needs answers all multiple choice (0-2)	Needs improvement in using concepts to design and propose solutions (0-3)	(1-10)